

Experiment No: 8

Title:

Access and validate form fields using events; design a gym admission form with live input checks.

Software/Tools Required:

JavaScript, HTML, CSS and Code Editor.

Theory:

Form Handling, Events, and Validation in JavaScript

Forms are an essential part of web applications as they allow users to enter and submit data. JavaScript provides powerful features to access form fields, handle user interactions through events, and validate input data before submission. Form validation ensures that users enter correct and complete information, improving data accuracy and user experience.

1. Form Fields in JavaScript

Form fields such as text boxes, radio buttons, checkboxes, dropdown menus, and password fields can be accessed using JavaScript through the Document Object Model (DOM).

Example:

```
let name = document.getElementById("name").value;
```

This statement retrieves the value entered in the input field with the ID name.

2. Events in JavaScript

An event is an action that occurs in the browser, such as a mouse click, key press, or form submission. JavaScript can respond to these events and perform specific actions.

Common form-related events include:

a) click Event

Triggered when a user clicks an element.

Example:

```
button.addEventListener("click", function() {  
    alert("Button Clicked");  
});
```

b) input Event

Triggered whenever the user enters or modifies data in a field.

Example:

```
input.addEventListener("input", function() {  
    console.log("Input Changed");  
});
```

c) change Event

Triggered when the value of a form field changes and loses focus.

Example:

```
select.addEventListener("change", function() {  
    console.log("Selection Changed");  
});
```

d) submit Event

Triggered when a form is submitted.

Example:

```
form.addEventListener("submit", function(event) {  
    event.preventDefault();  
});
```

3. Form Validation

Form validation ensures that user input satisfies specified conditions before submission.

Common validation checks include:

- Required fields should not be empty.
- Name should contain only alphabets.
- Age should be within an acceptable range.
- Email should follow a valid format.
- Mobile number should contain the correct number of digits.
- Passwords should meet security requirements.

Example:

```
if (name === "") {  
    alert("Name is required");  
}
```

4. Live Input Validation

Live validation checks user input in real time while data is being entered. This is commonly achieved using the input event. Immediate feedback helps users correct mistakes before form submission.

Benefits of Live Validation:

- Improves user experience.
- Reduces form submission errors.
- Provides instant feedback.
- Saves processing time.

Case Study:

A Gym Admission Form was created using HTML and CSS, and JavaScript was used to access and validate the form fields. Focus and blur events were implemented for user interaction, while name, gender, eye color, and athletic ability were validated before submission. Checkbox values were also handled, and appropriate error or success messages were displayed dynamically.

Output :

FORGE FITNESS / 2025

Build the strongest version of you.

A focused space, expert coaching, and a community that keeps you moving.

24/7
access for members

Start your journey

Tell us a little about yourself to get started.

Full name

Arpita Waghmare

Age

19

Phone number

8830385181

Email address

arpita.waghmare.batch2024@sitnagpur.siu.edu.in

Fitness goal

Build strength

Membership plan

Yearly - Rs. 11,999

☒ I agree to the membership terms and confirm that the details above are correct.

Join Forge Fitness →

Admission form submitted successfully!

FORGE FITNESS / MEMBER PROFILE

Gym Admission Form

NAME	VALUE
Name	Arpita Waghmare
Sex	☐ Male ☒ Female
Eye color	Brown
Check all that apply	☐ Over 6 feet tall ☒ Over 200 Pounds
Describe your athletic ability:	
I am active and enjoy doing regular exercise. I have good stamina and try to maintain my fitness	
Enter my information	

Thanks, Arpita Waghmare! Your female profile has been submitted.

Conclusion:

JavaScript enables efficient access to form fields and handling of user interactions through events. Form validation ensures that entered data is accurate and complete, while live input validation provides real-time feedback to users. A gym admission form demonstrates these concepts effectively by validating user details dynamically and ensuring a smooth registration process.